Towards quantisation of Boussinesq solitons: Master report through Phase 2i and the November 2026 packages

Project *quantum_boussinesq_solitons* — PI: Gustav Delius

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Abstract

(c) The strongest new result is the one-body fixed-momentum soliton coefficient $C_T = 1/(2\pi)$ in the stipulated Toda edge limit: an unconstrained numerical extraction supports it to about 5×10^{-5} relative precision, and a separate analytic calculation gives the same value. The formally continued continuum trace gives $2/\pi$ with different vacuum elastic renormalisation. Its sound-speed shift is $9/(4\pi)$, against Toda's $3/(4\pi)$. Matching vacuum elasticity alone fixes $\Delta\mathcal{H} = -3\hbar_B u^2/(2\pi)$ and changes the continuum coefficient to $1/(2\pi)$. The sound-subtracted slope is already $C - s_1 = -1/(4\pi)$ on both sides. These are conditional project results, checked by the one-body packages and `audit_onebody_scheme.py`; full quantum equivalence, a physical bad-equation determinant, and uniform interchange of limits remain **(d)**.

(c) The classical symplectic scattering phase survives the audits, including its potential-tail cross terms. The projected-determinant package establishes an instantaneous collective geometry and charge-pair invariance at fixed momentum, not a complete functional determinant. The Toda endpoint identities and the surviving mixed classical-frame remainder are confirmed. An arbitrary antisymmetric addition is not a physical frame ambiguity: cluster separability and the Krein equation exclude the displayed Toda counterexample. On the continuum side one antisymmetric overtaking function remains the residual $O(\hbar^0)$ two-body unknown, under the local analyticity and loop-expansion assumptions of the S-matrix audit. It is the same kind of functional obstruction, not an identified common function or a freely adjustable physical convention.

(c) Phase 2i supplies conditional formal integrals and orientation bookkeeping; its printed transforms, inverse series, reciprocity product and kernel necessity claim fail audit. Its strict two-loop accuracy and uniform finite-sea asymptotics are not established **(d)**. **(c)** The defect package supplies classical localized charges, a quadratic momentum trace and a boundary UV subtraction requirement; its universal fusion no-go is not proved. **(d)** No exact physical soliton S-matrix, absolute one-loop phase or quantum defect fusion rule is obtained.

Audit status and evidence precedence. **(c)** This report consolidates the coordinator's commit digests `a111f8c^..HEAD` through `66e4d35`, with each package subordinated to its independent audit. The earlier Phase 2h, charge/action, determinant, frames/defects, Toda-dictionary and projected-measure audits remain in force. The new controlling pairs are:

Package (all under `phase2/`)	Controlling audit
`toda_wh_and_kp_synthesis.tex`	`audit_phase2i.tex`
`frame_dictionary.tex`	`audit_frame_dictionary.tex`
`det_projected_oneloop.tex`	`audit_det_projected.tex`
`smx_candidates.tex`	`audit_smx.tex`
`defq_charges_fusion.tex`	`audit_defq.tex`
`onebody_dispersion_test.tex`, `onebody_toda_precision.tex`, `onebody_scheme.tex`	`audit_onebody.tex`, superseded where corrected by `audit_onebody_scheme.tex`

(c) The latest scheme audit refutes the earlier one-body audit's zero continuum sound shift and records its still-failing numerical assertion. Its analytic Toda coefficient and dictionary checks survive; its claim of uniform limit interchange is not adopted. `errata.tex`, `../scripts_inventory.md`, `../phase0/reference_audit.md` and `../paper/review_draft1.tex` supply additional corrections and coverage limits. No new equation, regularisation or physical phase convention is selected here. Labels mean literature proof (a), literature heuristic (b), project result (c), and unresolved interpretation or research judgment (d). Mathematical claims in the inherited classical derivation below are (c), except where separately labelled; their named checks are inherited evidence, not scripts rerun for this consolidation.

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1 Setting

(c) The initial PI-selected branch is the bad equation, with field $u_b = 2(\log\tau)_{xx}$ and $k = \sinh\beta$. A soliton has velocity $v = \epsilon\cosh\beta$, with $\epsilon = +1$ for right-movers and $\epsilon = -1$ for left-movers. The potential is $u_b = \phi_x$ and the Lagrangian is

$$\mathcal{L} = \tfrac{1}{2}\phi_t^2 - \tfrac{1}{2}\phi_x^2 + \tfrac{1}{2}\phi_{xx}^2 - \phi_x^3,$$

with $\pi = \phi_t$ and canonical symplectic form $\Omega = \int \delta\pi \wedge \delta\phi$.

On one soliton the charges are:

- mass $M = \int u_b\,dx = 2k$;
- ϕ-shift charge $\Pi = \int \pi = -vM$, conserved because $\phi \to \phi + c$ is a symmetry;
- momentum $P = -\int \pi\phi_x = \frac{2}{3}\epsilon k^3\cosh\beta$;
- energy $E = \frac{2}{15}k^3(5 + 4k^2)$.

These satisfy $dE/dP = v$ **(c)**. The corresponding good branch uses $k = \sin\alpha$, $v = \epsilon\cos\alpha$, cubic term $+\phi_x^3$ and dispersive term $-\phi_{xx}^2/2$ in the Lagrangian, and $E_g = 2k^3(5 - 4k^2)/15$. Checks: `errata_checks.py`, groups `energies` and `source_dictionaries`. For the BC field, $\varphi_{BC} = -\phi_b$; for CZ, $u_{CZ} = -3\phi_g$ and $\mathcal{L}_{CZ} = 9\mathcal{L}_g$. The corrected bad stable-band normal-ordering shift is $-3\hbar_B\sinh\beta/\pi$, not the historical positive sign. The old zero-scheme tables do not change (`errata.pdf`, M01–M03).

The two-soliton constants are

$$A_{\text{ov}} = \frac{1 + 3\coth^2\frac{\beta_1+\beta_2}{2}}{1 + 3\coth^2\frac{\beta_1-\beta_2}{2}}, \qquad A_{\text{ho}} = \frac{1 + 3\tanh^2\frac{\beta_1-\beta_2}{2}}{1 + 3\tanh^2\frac{\beta_1+\beta_2}{2}},$$

both in $(0, 1)$, and the position shifts are $\Delta x_1 = -k_1^{-1}\log A$, $\Delta x_2 = +k_2^{-1}\log A$ (soliton 1 starts on the left). **(c)** (Phase 1)

What a soliton S-matrix would mean here. **(c)** Neither chosen equation has the assumed relativistic boost symmetry; its long-wave vacuum dispersion is gapless. The bad vacuum also has unstable short-wave modes. The natural object is therefore the non-relativistic two-body S-matrix $S(P_1, P_2)$ of soliton asymptotic states, in the sense of Faddeev–Korepin [7]. **(c)** Regular classical two-soliton scattering is transmissive. **(d)** A unitary diagonal quantum soliton sector would therefore be a natural target, but is not constructed by that classical fact.

(c) For scalar diagonal scattering the Yang–Baxter equation is automatic. The non-trivial requirements are:

- unitarity;
- the correct semiclassical limit;
- analyticity and a crossing substitute;
- a bound-state (breather) spectrum consistent with poles.

2 Route A: semiclassical quantisation

2.1 The symplectic structure of soliton scattering

Proposition 2.1 (**(c)**). *Consider two well-separated solitons, with ϕ normalised by $\phi(-\infty) = c$. Up to exponentially small terms,*

$$\Omega = d\Pi_{\text{tot}} \wedge dc + \sum_j dP_j \wedge dX_j + \sum_{i\neq j} \big[X_i > X_j\big]\, d\Pi_i \wedge dM_j.$$

The argument is as follows. π_i is localised at X_i, and there ϕ_j equals 0 or M_j according to whether soliton i is to the left or right of soliton j. The self-term $dP \wedge dX$ was derived in Phase 1. In sine-Gordon the analogous cross term vanishes, because the kink height $2\pi/\beta$ is fixed. Here $M = 2k$ varies with the soliton parameter.

Proposition 2.2 (**(c)**, `scripts/symplectic_scattering.py`)**.** *The classical scattering map $X_j \mapsto X_j + \Delta x_j$ (with k_j and c fixed) preserves Ω, i.e.*

$$\sum_j dP_j \wedge d\Delta x_j = (\textit{cross term})_{\text{in}} - (\textit{cross term})_{\text{out}}.$$

This was verified to 40 digits for overtaking and head-on collisions at five parameter points each. It was also verified for the good *equation in its regular region (`scripts/consistency_checks.py`). Without the cross terms the identity fails by $O(1)$ amounts; for example, the left side equals* 19.79 *at $\beta = (0.9, 0.3)$.*

This is a non-trivial confirmation of the classical picture. It also shows that a semiclassical S-matrix built from the naive rule "$\Delta x_j = -\partial\delta/\partial P_j$" does not exist: the shifts are not curl-free in the canonical momenta.

2.2 The semiclassical phase

In each ordering region we choose Darboux coordinates by the rule

$$X_j' = X_j + \gamma_j \sum_{i \text{ left of } j} M_i, \qquad \gamma_j = \frac{d\Pi_j/dk_j}{dP_j/dk_j}.$$

The mirrored rule (normalising ϕ at $+\infty$) gives the same in–out differences. The shifts $\Delta x_j'$ are then curl-free. Their potential is the semiclassical phase δ, with $\Delta x_j' = -\partial\delta/\partial P_j$ and $S \simeq \exp(i\delta/\hbar)$, by stationary phase on two-particle wave packets [7].

Proposition 2.3 (**(c)**, checked semiclassical phase)**.** *1.* $\dfrac{dP}{d\beta} = \frac{2}{3}\,\epsilon k \sinh 3\beta$ *for the bad equation. For the good equation,* $\dfrac{dP}{d\alpha} = \frac{2}{3}\,\epsilon k \sin 3\alpha$ *with $k = \sin\alpha$; this vanishes at the cusp $\alpha = \pi/3$ found in Phase 1.*

2. For both overtaking and head-on collisions, and both equations,

$$\boxed{\delta(1,2) = \tfrac{2}{9}\,(\ell_1 - \ell_2)\,\log A_{12} + \tfrac{2}{3}\,M_1 M_2\,(v_1 - v_2)} \tag{1}$$

with $\ell = \epsilon\cosh 3\beta$ (bad) or $\ell = -\epsilon\cos 3\alpha$ (good). The remainder was first found by symbolic integration, then identified as $\frac{2}{3}M_1M_2(v_1 - v_2) = -\frac{2}{3}(\Pi_1 M_2 - \Pi_2 M_1)$ by exact matching.

3. Unitarity: $\delta(1,2) = -\delta(2,1)$.

4. Crossing-type relation: $\delta_{\text{ov}}(\beta_1, \beta_2 + i\pi) = -\delta_{\text{ho}}(\beta_1, \beta_2)$. *The shift $\beta \to \beta + i\pi$ maps a right-mover to a left-mover with $M \to -M$.*

Checks: `scripts/semiclassical_phase.py` and `scripts/consistency_checks.py`.

Remark 2.4.
- ℓ is proportional to the spectral parameter of the third-order Lax operator. In the uniformisation of [4], $\lambda = \frac{1}{2}(k^3 + k^{-3})$, with k^3 playing the role of $e^{3\beta}$.
- The dynamical part of δ is (difference of spectral parameters)$\times \log A$.
- The second part is a "mutual statistics" phase: an Aharonov–Bohm-like phase $\propto \Pi_1 M_2 - \Pi_2 M_1$ between objects carrying ϕ-shift charge and ϕ-winding. **(d)** (interpretation)

Remark 2.5 (KdV limit, **(c)**). For small amplitudes the topological term is of the *same order* as the KdV-type term $2\beta_1 \log A$ in $\partial\delta/\partial\beta_1$. At $\beta_1/\beta_2 = 1.7$, the full derivative divided by the KdV term tends to $0.564246498175\ldots$; the cross-only ratio is $-0.435753501825\ldots$ (`errata_checks.py`, `minor`). The literature heuristic **(b)** of Bonnemain and Doyon [2] identifies the quantum phase derivative with $P'_{\mathrm{GHD}}\Delta$, with $P_{\mathrm{GHD}} = \eta^2/2$. That identification reproduces our dynamical term but *misses the topological term.* The classical generalised hydrodynamics, which uses only Δx, is unaffected. **(c)**

2.3 Bohr–Sommerfeld: breathers as spectral-parameter strings

Bogdan and Charkina propose the semiclassical quantisation **(b)** of the Tajiri–Murakami breather of the bad equation by Bohr–Sommerfeld, $I_{\mathrm{br}} = N\hbar$, with an independently computed adiabatic invariant I_{br} **(a)** for the classical action, **(b)** for the large-quantum-number prescription [1, (64)–(65)].

Proposition 2.6 (**(c)**). *Set $\beta_{1,2} = a \pm ib$, which is Bogdan–Charkina's parametrisation $\kappa \pm ik = \sinh(a \pm ib)$. Then:*

1. *A_{12} equals their a_{12} [1, (33)]. In the stated localized window $0 < b < \pi/2$, regularity $A > 1$ holds precisely for $b > \pi/3$. The endpoint $b = \pi/2$ needs a separate localization condition (`audit_smx.pdf`).*
2. *$i \cdot \frac{2}{9}(\ell_1 - \ell_2) = I_{\mathrm{br}}$ exactly, with $I_{\mathrm{br}} = -\frac{4}{9}\sinh 3a \sin 3b$, in agreement with [1, (64)].*

Hence the coefficient of $\log A$ in (1) gives the classical breather action. Conditional on the semiclassical Bohr–Sommerfeld prescription,

$$\ell_1 - \ell_2 = -\tfrac{9i}{2} N\hbar, \qquad N \in \mathbb{N}, \tag{2}$$

which is a string condition in the spectral parameter ℓ.

This is the strongest cross-check in this report. It ties our two-soliton phase, built from Hirota shifts and the canonical symplectic form, to an independent quasiclassical calculation in the literature. Item 2 was checked at three parameter points to 10^{-16}.

Equation (2) is the form in which bound states appear in Bethe-ansatz solvable models. It suggests that ℓ is the variable in which an exact S-matrix has poles: $\ell_1 - \ell_2 \in -\frac{9i}{2}\hbar\mathbb{N}$, possibly with $\hbar$ renormalised. **(d)**

2.4 One-loop and candidate updates through November 2026

Continuum and projected determinant (Phases 2e, 2f, 2h). **(c)** Reflectionless tau-function phonon modes, dilogarithmic continuum integrals and direct potential-field spectral-density checks survive. The regulated co-moving trace requires its equal-mode-count cutoff endpoint and finite tadpole subtraction. The good-branch conditional fixed-momentum coefficient is $-2/\pi$ in that prescription. The new projected package proves

$$F|_P = e - vp = e_{\mathrm{can}} - vp_{\mathrm{can}} = -L_1, \qquad L_1 = vp - e,$$

where $E(P) = E_0(P) + \hbar_B F(P) + \cdots$. At fixed L_1, invariance between these two charge pairs is algebraic and unconditional; identifying L_1 with the complete effective action is not. Checks: `audit_det_projected.py`, checks 1–5, with its independent replacements for four tautological check groups.

(c) Translation, velocity and shift tangents have rank two. An auxiliary momentum direction gives a finite instantaneous projector and symplectic measure, but cannot remove the intrinsic kinetic-norm divergence caused by the velocity-dependent potential jump. The fixed-endpoint

fluctuation domain, constraint evolution, Schur complement and time-boundary determinant remain missing **(d)**. Thus neither the conditional good coefficient nor its bad analytic continuation is a completed physical determinant. The vacuum matching that now reconciles the leading continued one-body coefficient is in Route D.

(c) At fixed physical momentum the original mode-charge conversion has logarithmic small-amplitude terms; the fixed-amplitude phonon limit alone does not establish a quantum Toda match (`matching.pdf`). The mode/canonical charge choice cancels from $F|_P$ but can still affect two-body frame terms. **(d)** The absolute exchange/Maslov phase and the complete physical scattering amplitude remain unresolved.

Bad branch and candidate tests. **(c)** The real stable-band trace in `oneloop.pdf` differs from the formal continuation used in the one-body comparison. Its extra resonance is a co-moving group-velocity condition; discarding unstable modes is not a construction of a full quantum field theory **(d)**. The elastic matching term in Route D does not repair that stable-band trace.

(c) The Gamma algebra and candidate poles at the recorded bad-equation Bohr–Sommerfeld levels survive `audit_smx.pdf`; no $N = 0$ pole occurs. **(d)** These are not an exact quantum spectrum or a derived S-matrix. The zero-background two-phase good Hirota breather class is singular **(c)**; this does not exclude every possible quantum bound state or other background **(d)**. The printed good-breather threshold, extra resonance line and smoothness claim are refuted: the genuine resonance limits are continuous but generally not differentiably smooth (`audit_smx.py`).

Ring equations and the residual unknown. **(c)** The corrected Darboux wall signs, conditional directional ring derivative and total charge sum rule are checked by `audit_smx.py`. The printed total Bethe derivative omits a self-winding term. **(d)** The quantum seam holonomy is not derived from wavefunction transport, so the proposed twisted equations remain conditional. Under compatible local analytic branches, a nonzero amplitude ratio and an assumed loop expansion, one real antisymmetric overtaking *function* parametrises the residual $O(\hbar^0)$ two-body unknown; crossing then determines its continued head-on channel **(c)**. This is an unresolved physical calculation **(d)**, not a licence to choose an arbitrary physical frame. The relation to the Toda constraints is in Route D. The audit refutes the printed J crossing offset and Stirling cancellation: direct quadrature gives crossing sum zero, not $\pi/2$. It also refutes the printed small-amplitude interface. None is used here to fix a physical constant.

3 Route B: quantum inverse scattering on the line

(d) *What would have to be true.* The Lax matrix $U(x, \lambda)$ would need an ultralocal r-matrix Poisson algebra, so that it can be regularised on a lattice and quantised à la Faddeev–Korepin–Sklyanin. Then a Bethe ansatz over a pseudo-vacuum with soliton excitations would give S.

Proposition 3.1 (**(c)**, `scripts/routes_BC_checks.py`)**.** *In canonical variables the Lax data of Phase 1 are* $q = \frac{3}{2}\phi_x - \frac{1}{4}$ *and* $r = \frac{1}{2}q_x + \frac{\sqrt{3}}{4}\pi$ *(good-equation real form; the bad one differs by* i*'s). Their brackets are*

$$\{q(x), r(y)\} = \tfrac{3\sqrt{3}}{8}\,\delta'(x-y), \qquad \{q, q\} = \{r, r\} = 0.$$

So the companion-matrix Lax connection is non-ultralocal.

Where it stalls. There are two known ways out, and neither yet gives a theory on the line.

1. Maillet's r/s formalism with lattice regularisation [14]. No such line-soliton construction has been established in this project.

2. The Miura (free-field) map. This is the route of Bazhanov–Hibberd–Khoroshkin [3] and Hikami [10]. It yields $U_q(\widehat{\mathfrak{sl}}_3)$ T- and Q-operators, but in the chiral conformal setting on a circle, with no particle sector.

Dimakis et al.'s Yang–Baxter maps [6] concern polarisations in the matrix case and are vacuous for the scalar equation. The "beyond Yang–Baxter" remark there is therefore not an obstruction here. For diagonal scattering, Yang–Baxter holds trivially. **(c)**

4 Route C: W_3, sl_3 affine Toda and the $U_q(\widehat{\mathfrak{sl}}_3)$ R-matrix

(d) *What would have to be true.* Boussinesq solitons would have to be identifiable as a limit of $a_2^{(1)}$ (complex) affine Toda solitons, whose S-matrices are the $U_q(\widehat{\mathfrak{sl}}_3)$ intertwiners of Hollowood [12] and Gandenberger [8].

Proposition 4.1 (**(c)**, `scripts/routes_BC_checks.py`)**.** *Take Hollowood's $a_2^{(1)}$ interaction constant [11, (1.19)], $A^{(12)}(\theta) = \frac{2\cosh\theta+1}{2(1+\cosh\theta)}$, between the two fundamental species. With $S = \beta_1+\beta_2$ and $D = \beta_1-\beta_2$,*

$$A_{\mathrm{ov}}^{\mathrm{bad}}(\beta_1,\beta_2) = \frac{A^{(12)}(S)}{A^{(12)}(D)}\cdot\frac{\tanh^2(D/2)}{\tanh^2(S/2)}, \qquad A_{\mathrm{ho}}^{\mathrm{bad}}(\beta_1,\beta_2) = \frac{1}{A_{\mathrm{ov}}^{\mathrm{bad}}(\beta_1,\beta_2+i\pi)}.$$

Here $\tanh^2(\theta/2)$ is the sine-Gordon soliton–soliton constant. Equivalently,

$$A_{\mathrm{ov}} = \frac{G(S/2)}{G(D/2)}, \qquad G(y) = \frac{16\prod_{a=1,2}\sinh\frac{y+ia\pi/3}{2}\sinh\frac{y-ia\pi/3}{2}}{\sinh^2 y}.$$

So the Boussinesq interaction is built from exactly the $\mathfrak{sl}_3$ blocks, but in the form "$F(\beta_1-\beta_2)/F(\beta_1+\beta_2)$". This is the structure of scattering off a particle *and its mirror image*. It suggests that a Boussinesq S-matrix might take the form

$$S(\beta_1,\beta_2) = \mathcal{S}(\beta_1-\beta_2)\,\mathcal{S}(\beta_1+\beta_2+i\pi)^{\pm1}$$

with $\mathcal{S}$ built from $U_q(\widehat{\mathfrak{sl}}_3)$ and sine-Gordon factors. Such a form is reminiscent of boundary (reflection) amplitudes. That would connect to the "boundary affine Toda" structure in Bazhanov–Hibberd–Khoroshkin [3]. **(d)**

(c) The checked obstruction in the chosen rapidities is that the semiclassical phase (1) is not of this product form. Its weight $\frac{2}{9}(\ell_1-\ell_2)$ is not a sum of functions of S and D, and its topological part has no Toda analogue.

Further obstructions:

- Toda solitons are relativistic with fixed masses m_a. In the Toda constant the mass term $4m^2\sin^2(\pi a/3)$ appears where Boussinesq has K^4, so the species label is "replaced" by the wavenumber. No such limit or quantum dictionary has been constructed here; the displayed dispersion difference is not a universal impossibility proof **(d)** (`../paper/review_draft1.tex`, R12).

- Takács and Watts show that $a_2^{(1)}$ soliton–breather scattering is flawed both classically and quantum mechanically **(b)** [20].

(d) *Route C remains open*: the structural similarity is precise, but no dictionary is available. What is lost in the non-relativistic reduction is the difference form, and with it the standard crossing.

November candidate audit. **(c)** The classical block factorisation remains valid. The universal exclusion of mixed-argument quantum dilogarithms is refuted; a repaired residue argument excludes the specific nine KP-root ratios with constant coefficients. Exchange unitarity excludes a nonconstant smooth mirror factor, but permits a constant mirror factor and an exchange-unitary difference factor. The separate classical nonseparability test is not an exclusion of all quantum products. Checks: `audit_smx.py`, `families`. **(d)** No new affine-Toda quantum identification follows from these restricted exclusions.

5 Route D: Toda edge data, one-body matching and the two-body problem

Classical and finite-sea foundations (Phases 2b–2d). **(a)/(b)** The inherited quantum Toda and asymptotic Bethe framework is described in [9, 17, 19, 15, 18, 16]. **(c)** The implemented classical limit reproduces the Toda soliton phase with topological terms; its small-amplitude scaling recovers the bad-equation classical phase, with $\hbar_T = 3b\hbar_B$, $b = e^{-a/2}$ and $\kappa \sim \sqrt{3}\beta$. Checks: `toda_bethe_classical.py`, `toda_ba_vs_label.py`, `toda_scaling_fast.py`; the six dictionary normalisations are confirmed by the one-body scheme audit. This is a small-amplitude correspondence, not finite-amplitude quantum equivalence **(d)**.

(c) The Phase 2d head-on agreement and overtaking discrepancy from the Gamma candidate are results in the implemented classical frame. The claimed bound on successive raw phase changes, < 0.004, fails: the recorded changes are 0.006326, 0.009016, 0.004512 (`../scripts_inventory.md`). The extrapolated table checks pass separately. **(d)** Neither establishes a physical quantum S-matrix without the common continuum map.

One-body coefficient: the current main result. **(c)** Define F by $E(P) = E_0(P) + \hbar_B F(P) + \cdots$ and $C = \lim_{k\to 0} F(k)/|P_0(k)|$ within the specified edge/semiclassical extraction. This energy correction contributes at $O(\hbar^0)$ to the semiclassical phase. The initial `onebody_dispersion_test.pdf` found a factor-of-four mismatch. The higher-precision package now gives

$$C_T^{\mathrm{num}} = 0.1591549414 \ \pm \ 8.24 \times 10^{-6}, \qquad C_T = \frac{3}{4\pi} - \frac{1}{4\pi} = \frac{1}{2\pi}.$$

The rounded-up envelope combines mesh/roundoff 5.68×10^{-8}, finite-$\hbar_T$ extrapolation 7.70×10^{-6}, and amplitude-tail 4.73×10^{-7} diagnostics. It is deterministic fit sensitivity, not a statistical confidence interval or rigorous error bound; the relative scale is about 5×10^{-5}. The unconstrained fit does not insert $1/(2\pi)$. The separate analytic calculation uses the exact Bethe derivative identity, momentum logarithm and harmonic vacuum normalisation (`onebody_toda_precision.pdf`, `onebody_prec_identities.py`, `onebody_prec_edge.py`; `audit_onebody_scheme.pdf`). Uniform interchange with fixed nonzero momentum remains **(d)**, despite the stronger wording of the earlier one-body audit.

Vacuum elasticity resolves the leading mismatch. **(c)** In the inherited prescriptions, write $c_{\mathrm{sound}} = 1 + \hbar_B s_1 + \cdots$. The continued bad continuum and Toda give

$$C_{\mathrm{ac}} = \frac{2}{\pi}, \quad s_{1,\mathrm{ac}} = \frac{9}{4\pi}, \qquad C_T = \frac{1}{2\pi}, \quad s_{1,T} = \frac{3}{4\pi}.$$

Matching their vacuum elastic moduli, with no soliton target used, fixes the Hamiltonian density correction

$$\Delta\mathcal{H} = \hbar_B \lambda u^2, \qquad \lambda = s_{1,T} - s_{1,\mathrm{ac}} = -\frac{3}{2\pi}.$$

Since $P_0 = v\int u_0^2 dx$ and $|v| \to 1$, this shifts both C and s_1 by λ. Hence

$$C_{\text{matched}} = \frac{2}{\pi} - \frac{3}{2\pi} = \frac{1}{2\pi}, \qquad C_{\text{ac}} - s_{1,\text{ac}} = C_T - s_{1,T} = -\frac{1}{4\pi}.$$

Checks: `onebody_scheme_continuum.py`, `audit_onebody_scheme.py`, and the read-only consolidation check `errata_checks.py`, `november_onebody`. Scheme invariance here means invariance under the common finite elastic subtraction, with the inherited origins/tadpole fixed. **(d)** The physical bad determinant, complete quantum equivalence, finite-amplitude matching and uniform limits are still open. The real stable-band resonance is not removed by this counterterm. For the good equation the conditional bare values are $C_g = -2/\pi$ and $s_{1,g} = -9/(4\pi)$, giving $C_g - s_{1,g} = +1/(4\pi)$ **(c)**; Toda forces no good-branch prescription **(d)**.

Frame dictionary: what is now fixed. **(c)** `audit_frame_dictionary.pdf` confirms the exact finite-sea endpoint, Jacobian and pressure-response identities, independently checks derivatives missing in the package, and confirms survival of the mixed classical-frame remainder in the edge limit. Its small-y error budgets require conditioning qualifications. One-body phase-reference shifts cannot cancel the mixed remainder. The exhibited antisymmetric addition evades only the incomplete one-body/reciprocity constraints; cluster separability and the microscopic Krein PDE exclude it as a physical Toda frame ambiguity. The parity/pole checks supply further restrictions (`audit_frame_dictionary.py`).

(d) What remains is to identify the allowed one-body endpoint-response bilinears with continuum collective operators and the physical two-body phase. The continuum's single residual antisymmetric function in `audit_smx.pdf` is the same *kind* of two-body unknown, not an established equality to the rejected Toda example. Neither side licenses that example as a physical freedom on its own terms. One-body agreement does not determine the function.

Phase 2i: conditional asymptotics, not completed reconciliation. **(c)** The controlling `audit_phase2i.pdf` refutes the printed Fourier symbol, momentum transform/scale, inverse series, double transform, four-term reciprocity product, Darboux coefficient and kernel "if and only if". With the missing normalisation restored, reversion and inverse-Laplace tests support the proposed edge coefficients. The formal integral identity

$$\frac{4}{\pi}\int_0^r \log(s/r)\frac{s^2}{1-s^2}\,ds = -\frac{4}{\pi}(\chi_2(r) - r), \qquad 0 < r < 1,$$

with $\chi_2(r) = [\mathrm{Li}_2(r) - \mathrm{Li}_2(-r)]/2$, and the proposed two-loop coefficient integral are confirmed as conditional calculations, not uniform Bethe asymptotics. Checks: `audit_phase2i.py`, `errata_checks.py:formal_integrals`. The corrected Krein convolution has independent numerical support. **(d)** Its finite-sea derivation still lacks zero-frequency matching and uniform endpoint control. The advertised $< 2 \times 10^{-4}$ data match uses one fixed-leading-coefficient estimator; alternative fits exceed it and can reverse subleading signs **(c)**. The earlier logarithmic remainder issue is therefore not settled. Phase 2i does not resolve all six earlier audits, derive a physical all-loop kernel, or fix breather/fusion constants **(d)**.

Other discrete route. **(a)** Hikami's periodic quantum discrete Boussinesq construction [10] provides a separate algebraic starting point. **(d)** A line-soliton ground state, particle sector and thermodynamic scattering dictionary have not been constructed here; this remains a longer-term alternative.

6 Route E: defects, boundary charges and qualified fusion

(a) CZ construct type-I defects for the good equation [5]. **(c)** In project units $u_{CZ} = -3\phi_g$, $\mathcal{L}_{CZ} = 9\mathcal{L}_g$. The scattering root uses the unsigned vacuum parameter:

$$s = \sigma_{\mathrm{vac}}, \qquad q_D = -\frac{s}{2\sqrt{3}}, \qquad z = \left(\frac{v + s - \sqrt{3}k}{v + s + \sqrt{3}k}\right)^{\eta} = \left(\frac{p_b - q_D}{p_a - q_D}\right)^{\eta}.$$

The spectrum instead uses $c = \eta\sigma_{\mathrm{vac}}$. These must not be interchanged at $\eta = -1$. Checks: `errata_checks.py:roots_defect` and `audit_defq.py`. **(c)** The classical phase and regular transmission factorisation survive in this dictionary (`defect_semiclassics.pdf`); reflection and two-soliton decay were not covered.

New localized-charge and trace results. **(c)** `audit_defq.pdf` independently confirms the four classical localized charges and their conservation using the bulk equations and sewing conditions, and KP-edge factorisation of their vacuum jumps. The package's claimed numerical absorption test accidentally tests transmission because of an extra η; genuine absorption checks in both orientations pass in the audit. The quadratic boundary momentum trace is confirmed by independently solved sewing modes. The cutoff Casimir trace has a linear UV divergence requiring boundary subtraction; a bulk tadpole on the open half-lines cannot remove it. **(d)** The finite boundary prescription, complete mode/state normalisation and quantum Ward identities are not established. The restricted counterterm test is not an exhaustive classification of quantum boundary operators.

Spectrum and limits on the fusion obstructions. **(c)** The corrected determinant prefactor is sixteen times the historical one, and the admissible growing branch switches at $c = \sqrt{3}$. The remaining positive rate there is $\sqrt{3}/4$. For three distinct real KP roots the counts are generically $(0, 1, 2)$ and $(0, 1, 1)$ at the switch; no edge joins two vertices free of growing modes in this analysis. Thresholds and repeated roots require separate treatment. Checks: `audit_defq.py`, `errata_checks.py:defect_spectrum` and `november_defect`. **(d)** This is not a nonlinear stability/completeness theorem or an exclusion of a separately constructed resonance continuation.

(c) Nonzero canonical-charge holonomies and absorbed-momentum remainders are confirmed at the sampled triangles. A single-valued vertex counterterm cannot cancel a demonstrated nonzero fixed-edge holonomy, by telescoping. **(d)** Universal nonvanishing, unavoidable physical obstruction, and a general quantum-fusion no-go do not follow: roots, rapidities, the charge dictionary or the state space might change. The package's pole argument against root shifts is refuted because its alleged pole has zero residue **(c)**. The printed energy-trace sign, negative-wavenumber phase limit and threshold rate are also corrected by the audit; details are in the errata addendum.

KP bookkeeping (Phases 2g–2i). **(c)** Abstract chord closure, dipole differences and explicit orientation constants survive independent quadrature. Phase 2i explains the earlier orientation offsets algebraically, by a redefined kernel; it does not make them disappear from the raw integrals. Inversion antisymmetry suffices for exchange of the defined ratio bilinears, but the claimed necessity is false. **(d)** Branchwise analytic continuation is distinct from the real-part kernel and does not prove a global physical breather amplitude. Absolute phases, normalized defect states, quantum sewing and physical fusion remain open.

7 Consistency checks across candidates and orders

Claim	Audited status and check	Label
Classical phase	Symplectic/Darboux phase survives; corrected bad action and KdV ratio in `errata_checks.py`.	**(c)**
One-body Toda	$C_T = 1/(2\pi)$, analytic and numerical to about 5×10^{-5} relative; `onebody_prec_identities.py`, precision package.	**(c)**
Vacuum matching	$9/(4\pi)$ versus $3/(4\pi)$ fixes elastic counterterm; $C - s_1 = -1/(4\pi)$ agrees; `audit_onebody_scheme.py`.	Conditional **(c)**
Projected geometry	Finite instantaneous measure; intrinsic kinetic tails. $F\|_P = -L_1$ charge-pair invariant; trace/determinant identification conditional; `audit_det_projected.py`.	**(c)**/**(d)**
Toda frame	Endpoint/Jacobian identities and mixed remainder confirmed; arbitrary antisymmetric example excluded by cluster/Krein constraints; `audit_frame_dictionary.py`.	**(c)**
Continuum two-body	One residual antisymmetric function under local analytic and loop assumptions; common physical map not derived; `audit_smx.py`.	**(c)**/**(d)**
Phase 2i	Corrected-transform/formal-integral evidence survives; printed identities refuted, uniform asymptotics and strict fit error unproved; `audit_phase2i.py`.	**(c)**/**(d)**
Candidates/ring	Gamma algebra and bad candidate poles pass; restricted exclusions only; ring identities conditional, quantum holonomy unproved; `audit_smx.py`.	**(c)**/**(d)**
Defect charges	Classical charges, momentum trace, UV subtraction and growing counts pass; printed sign/phase/threshold/pole argument fail; `audit_defq.py`.	**(c)**
Physical fusion	Sampled holonomy obstruction is conditional; no universal no-go or normalized quantum bootstrap.	**(d)**
Verification record	Phase 2d step bound and one-body audit tolerance still fail; tautologies replaced in audits, disabled Hirota check repaired; `../scripts_inventory.md`, errata addendum.	**(c)**
Full quantum theory	Physical bad determinant, exact S-matrix, full equivalence and uniform limits remain open.	**(d)**

8 Conclusions and ranked next steps

(c) The leading one-body comparison is now reconciled by independently computed vacuum elasticity, with an invariant sound-subtracted coefficient. The factor of four is not a missing dictionary factor. **(d)** This is a conditional leading-coefficient result, not a proof of full quantum equivalence. **(c)** Classical scattering, finite-sea identities and boundary-charge algebra remain useful foundations; the November audits substantially narrow their claimed quantum consequences. **(d)** The physical two-body phase of the *good* equation is the main unsolved observable, and no candidate is promoted to an exact S-matrix.

Ranked next steps (research judgments (d); re-ranked by the PI, 4 November 2026). The good (well-posed) equation is the main thread. The quantum Toda chain reaches only the *bad* equation in its continuum limit, and the Toda-matched counterterm was traced to the bad equation's short-wave band near its unstable region $|q| > 1$. Toda and bad-equation work therefore serves as validation of the method, not as the target.

1. **Fix the good equation's one-loop renormalisation condition.** Candidates: a principled acoustic condition (the physical sound speed or compressibility of the good vacuum); a regularisation from an integrable good-equation lattice (the quantum discrete Boussinesq/W_3 chain of Hikami and Belov–Chaltikian [10]); and a test of whether the

Toda-matched $\Delta\mathcal{H} = -(3\hbar_B/2\pi)u^2$ continues through the good/bad dictionary, which is expected to fail because the factor of four came from the bad equation's unstable band. (Package `good_renorm`.)

2. **Compute the good-equation two-soliton $O(\hbar^0)$ phase, including its antisymmetric part,** at fixed physical momenta in both channels, in the scheme fixed by item 1. Complete the boundary-compatible projected determinant as far as this requires. Test whether three-soliton cluster separability fixes the antisymmetric part, as it does on the Toda side. (Package `good_twobody`.)

3. **Determine the quantum fate of the good equation's resonant and number-changing channels** (Y-junctions, merging solitons, singular breathers): are their conservation laws compatible with fusion, and is the quantum soliton S-matrix elastic and factorised at all? (Package `resonant_channels`.)

4. **Construct the renormalised quantum defect boundary theory** (good equation only). Build on the localised charges and momentum trace; derive a boundary renormalisation condition, admissible states and quantum absorption shifts between stable vacua; prove or refute the holonomy obstruction for all KP triangles. (Package `defb_boundary`.)

5. **Return to candidate selection and ring quantisation for the good equation** after items 1–3: quantum seam holonomy, physical pole and branch data, and exclusions restricted to proved families.

Supporting validation (bad equation and quantum Toda).

V1. **Two-body test of the method.** Compare the uniquely determined Toda $O(\hbar^0)$ two-body phase at fixed dressed momenta with the bad-equation continuum two-soliton phase in the Toda-matched scheme. (Packages `twobody_toda`, `twobody_cont`.)

V2. **Uniform Toda edge and small-amplitude limits.** Test the interchange of limits at fixed P; refit the two-loop data with separate mesh, edge and amplitude errors; retain the failed bounds in the record. (Package `uniform_limits`.)

V3. **A physical bad-equation determinant prescription**, to be derived or explicitly chosen by the PI before the continued trace is extended.

A Phase 2 Write-ups and Verification Scripts

This revision changes the evidence status of the quantum synthesis; it does not introduce new external references or claim to have reread the inherited bibliography. Literature-reading scope is recorded in the independent notes. Write-ups and checkers are in `phase2/` and `phase2/scripts/`, with defect scripts also in `corrigan-zambon/`. Success markers apply only after actual assertions pass; generators and the two known failures are listed in `../scripts_inventory.md` and the errata addendum. No blanket success claim is made. This consolidation runs only `errata_checks.py`; legacy scripts that may rewrite caches are not rerun.

- **Phase 2a (`report.pdf`):** `scripts/symplectic_scattering.py`, `scripts/semiclassical_phase.py`, `scripts/fit_R.py`, `scripts/consistency_checks.py`, `scripts/routes_BC_checks.py`.
- **Phase 2b (`toda_route.pdf`):** `scripts/toda_bethe_classical.py`, `scripts/toda_bethe_quantum.py`, `scripts/toda_ba_vs_label.py`, `scripts/toda_scaling_fast.py`, `scripts/toda_quantum_corrections.py`.

- **Phase 2c** (`smatrix_ansatz.pdf`, `defect_semiclassics.pdf`): `scripts/smatrix_rejected_forms.py`, `scripts/smatrix_semiclassical_unitarity_crossing.py`, `scripts/smatrix_poles_bootstrap.py`, `scripts/defect_smatrix_good_continuation.py`, `corrigan-zambon/defect_*.py`.
- **Phase 2d** (`toda_edge.pdf`, `oneloop.pdf`): `scripts/toda_edge_ba.py`, `scripts/toda_edge_twoterm.py`, `scripts/oneloop_step0_wellposedness.py`, `scripts/oneloop_phase_shifts.py`, `scripts/oneloop_soliton_energy.py`, `scripts/oneloop_breather.py`.
- **Phase 2e** (`oneloop_phase.pdf`): `scripts/oneloop_good_phonon_scattering.py`, `scripts/oneloop_good_twosoliton_phase.py`, `scripts/oneloop_good_limits_cusp_resonance.py`.
- **Phase 2f** (`matching.pdf`): `scripts/matching_conventions.py`, `scripts/matching_toda.py`, `scripts/matching_analysis.py`.
- **Phase 2g** (`oneloop_defect_phase.pdf`): `scripts/oneloop_defect_scattering_and_stability.py`, `scripts/oneloop_defect_phase_and_fusion.py`.
- **Phase 2h** (`charge_frame_master.pdf`): `scripts/charge_and_frame_audit.py`, `scripts/kp_master_and_toda_edge.py`.
- **Independent audits:** `independent_phase2h_audit.pdf`, `audit_charges_action.pdf`, `audit_determinant_foundations.pdf`, and `audit_frames_defects.pdf`, with the corresponding `audit_*.py` scripts listed in the consistency table.
- **Follow-up audits:** `audit_projected_measure.pdf` and `audit_toda_dictionary.pdf`, checked by the corresponding named scripts and saved JSON results.

November evidence index. **(c)** The package/audit table above indexes all November notes. Their checkers are `audit_phase2i.py`, `audit_frame_dictionary.py`, `audit_det_projected.py`, `audit_smx.py`, `audit_defq.py` and `audit_onebody_scheme.py`. The historical `audit_onebody.py` has a known unchanged failure and is not certified by this index. Precision evidence comes from the `onebody_prec_*` scripts; vacuum matching comes from `onebody_scheme_*`. `errata_checks.py` adds read-only checks of the newly consolidated corrections without rerunning those packages.

November failures recorded. **(c)** The Phase 2d 0.004 step bound fails at three recorded steps; the earlier one-body audit's 3×10^{-6} tolerance fails with residual $2.2546734882 \times 10^{-4}$. Finer extrapolation and independent inversion support the limiting identity, but do not make the unchanged script pass. Four `det_*` tautology groups and one `frame_identities.py` tautology were replaced by independent checks in their audits. The disabled Hirota `orTrue` assertion was repaired in the hygiene package. The Phase 2i, S-matrix and defect refutations, including sign/branch errors and tests that passed without testing their printed claims, are itemised in `errata.pdf`. Failed audit-development checks remain in the individual audit notes; none is erased by this consolidation.

Failures along the way (Phase 2a; later phases record their own failures in each write-up).

- The direct symbolic simplification of the symplectic identity timed out, so we switched to 40-digit numerics.
- The good-equation phase first used the wrong sign of ℓ, and the test failed until it was corrected.

- A tolerance set below double precision caused spurious failures.
- The KdV-limit comparison with Bonnemain–Doyon *does not* agree. This is recorded as a finding, not a passed test.

Failures recorded by the independent quantum audits.

- Initial PDF builds had overfull filename lines and a temporary text-group scoping error; line breaking and group closure were repaired before the final rendered-page inspection.
- During an earlier report revision, the Phase 2h audit's source-snapshot guard failed because the report was edited while the script was running. The guard was retained; verification was repeated after editing stopped.
- The classical momentum derivatives in this report omitted the orientation factor ϵ and therefore applied only to right-movers. Both signed identities are now checked in `audit_projected_measure.py`.
- Raw-phase antisymmetry, zero diagonal, and orientation-independent master identification failed independent continuum checks.
- The displayed defect determinant prefactor and the complete growing-mode pair failed checks away from zeros and between the old sampled parameters.
- The proposed Toda residual was used to define part of its purported derivation; saved extrapolation alternatives changed the subleading sign. The claimed edge-reference explanation of a ratio-dependent slope also failed.
- Potential amplitude modes have nonlocal tails, invalidating the stated localization-only cancellation argument. Equal endpoint frequencies do not by themselves establish localized determinant cancellation.
- Profile, weighted-sum, and derivative-sign errors in intermediate charge formulas were found although the final conditional charge derivative passed. Boundary counterterms and localized defect momentum were not accounted for in the claimed physical absorption argument.

Development failures, repairs, precision, and the scope of each successful `ALL CHECKS PASSED` run are recorded in the audit notes. A successful algebraic script does not certify the unresolved physical identifications.

Reference verification and reading scope. **(c)** Entries below are inherited sources, with metadata corrections taken from `errata.pdf` and `../phase0/reference_audit.md`. No external full text or abstract was newly read for this consolidation. The errata records relevant full-text passages read for CZ and BC, abstract/metadata-only verification for many other entries, and access limits; those distinctions are retained here. Literature labels describe the inherited source scope, not a fresh literature proof. In particular the BC action is classical (a), its quantisation prescription semiclassical (b), and exact quantum pole identification (d).

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